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# A PALEOMAGNETIC STUDY OF SECULAR VARIATION IN NEW ZEALAND *

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Ancient secular variation in New Zealand was determined from paleomagnetic measurements on 22 volcanic formations with ages of less than 0.68 m.y. The angular standard deviation from the field of an axial dipole is 13.2° with 95% confidence limits between 10.9° and 16.7°. The angular standard deviation of the corresponding virtual geomagnetic poles is 19.6° with confidence limits between 16.2° and 24.7°. These values are larger than those predicted by most models for secular variation. No difference was detected between the angular secular variation in New Zealand and that at the same latitude in North America.

## 1. INTRODUCTION

At the present time, both the nondipole field and the rate of geomagnetic secular variation (SV) are greater in the southern hemisphere than in the northern [1, 2]. Considerable theoretical interest attaches to the question of whether this condition is permanent or transient. The geomagnetic field is produced by dynamo processes in the earth's core [3]. If the hemispherical asymmetry in SV is permanent, it points to a long-term asymmetry in the boundary conditions at the core-mantle interface that govern the magnetohydrodynamic processes. Conversely, if the asymmetry between the hemisphere is transient and vanishes on taking long-time averages of the nondipole field, then no long-term asymmetry in boundary conditions is implied.

Over how long a time interval must measurements be made in order to assess the total variance of the geomagnetic SV field? Several lines of evidence indicate that the spectrum extends to periods of $10^4$ years [2, 4-6], so that measurements over intervals several times longer than this are needed. Such data are available in the paleomagnetic record contained in rocks. The most readily obtained information is the angular dispersion of the field at a number of discrete points of past time. This dispersion is determined by two factors. One is the angular wobble of the main dipole about the earth's rotation axis. The second is the average ratio $f/F$ of the nondipole field to the dipole field. The former produces a zonal angular dispersion that is symmetrical in both hemispheres. The latter may or may not produce asymmetry in the two hemispheres, depending on whether the average value of the ratio of $f/F$ is symmetrical about the equator. Aside from the question of the asymmetry of the hemispheres, variations with latitude of the quantity $f/F$ are of interest in testing models describing SV [2, 7, 8].

The time interval selected for the present study was from 0.68 m.y. ago to the present. This corresponds to the Brunhes normal polarity epoch. It is a particularly convenient interval because most of the rocks of this interval, including the oldest ones, are normally magnetized, whereas those of the older Matuyama epoch are reversely magnetized. Thus outcrops of suitable age can be easily identified without the necessity of radiometric dating. The present study provides data from the southern hemisphere to

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compare with the numerous data from this same interval already available from the northern hemisphere. In a previous study, Chamalaun [9] found rather large angular dispersion for the directions of magnetization of lava flows from Réunion Island in the Indian Ocean. However, because of the sampling scheme that was used, the observed angular dispersion cannot be attributed entirely to ancient geomagnetic secular variation.

## 2. GEOLOGIC AGES OF THE ROCKS SAMPLED

Two of the volcanic formations sampled on North Island are Holocene (post-glacial). The others are from the Wanganui Series (early and middle Pleistocene) and the Hawera Series (late Pleistocene). The geology and stratigraphy of these rocks are well known from the studies of Grindley [10], Healy, Schofield and Thompson [11], and Thompson and Kermode [12]. The first step in the present study was to determine the point in the stratigraphic sequence corresponding to 0.68 m.y., the age of the beginning of the Brunhes normal polarity epoch. This was done by measuring the magnetic polarity of rock samples with a portable fluxgate magnetometer [13]. It was found that the post-glacial volcanic units NZ-17 and NZ-28 (fig. 1) were normally magnetized, as were all of the volcanic units of the Hawera Series (fig. 1 and table 1). The boundary between normally magnetized units of the Brunhes polarity epoch and reversely magnetized units of the Matuyama epoch was found to occur within the sequence of rhyolite ignimbrites that fill the Taupo graben. These reach a maximum thickness of 1000 meters and cover an area of 36 000 $km^2$ [14]. All are normally magnetized except for two of the oldest ignimbrites exposed on the west flank of the graben. The reversed-to-normal transition between the Matuyama and Brunhes polarity epochs to 0.68 m.y. occurs between the Waiteariki ignimbrite (site NZ-8) and the Te Kopia ignimbrite (site NZ-14). These are members of the Wanganui Series and are two of the oldest ignimbrites in the sequence. There is no geologic evidence for a long hiatus between these units, suggesting that ignimbrite eruptions began slightly before 0.68 m.y. years ago.

At first glance this paleomagnetic stratigraphy appears to be inconsistent with earlier attempts to date radiometrically the beginning of the Pleistocene in New Zealand. This work indicated that deposition of the Wanganui Series, generally regarded as the beginning of the Pleistocene, commenced 2.5 to 2.1 m.y. ago [15-17]. The ignimbrites which record the normal-to-reversed polarity transition were regarded by Healy et al. [11] as correlative with the lower part of the Wanganui Series and hence early Pleistocene, although they stress the uncertainty of the correlation. The inferred age of the earlier ignimbrites would thus be about 2 million years, whereas the present interpretation is that they are about 0.7 m.y. old. The simplest explanation is that the earliest part of the ignimbrite sequence is not the time equivalent of the lower Wanganui Series at the localities where the latter were dated radiometrically, but is considerably younger.

Support for this interpretation is provided by the paleomagnetic stratigraphy of marine sediment cores studied by Ninkovich [18] east and northeast of North Island. The sediment cores contain five distinct layers of windborne rhyolitic ash probably correlative with the ignimbrite eruptions. The oldest two ash layers are reversely magnetized and the youngest three are normal. Assuming deposition at a uniform rate and an age of 0.7 m.y. for the Matuyama-Brunhes boundary, Ninkovich [18] found ages of 0.85, 0.73, 0.67, 0.31, and 0.27 m.y. for the five layers. Ash layers are absent from the lower sections of the cores, suggesting that no large ignimbrite eruptions occurred between 2.5 and 0.85 m.y. ago. The paleomagnetic stratigraphy on land and that in the marine sediments thus agree. Apparently, the rhyolite eruptions began slightly before the Matuyama-Brunhes boundary set at 0.68 m.y. ago.

All of the volcanic formations previously mapped as being younger than the Waiteariki ignimbrite (site NZ-8) were found to be normally magnetized. The only exception is the Omahia andesite (site NZ-6), which is reversely magnetized and hence probably somewhat older than the Te Kopia ignimbrite – exactly how much older is not known. Otherwise the paleomagnetic and geologic results are in good agreement. However, the North Cape stratigraphy is less well known than that of the central volcanic zone, and the possibility cannot be precluded that units NZ-1 and NZ-4 are from normal epochs older than the Brunhes.

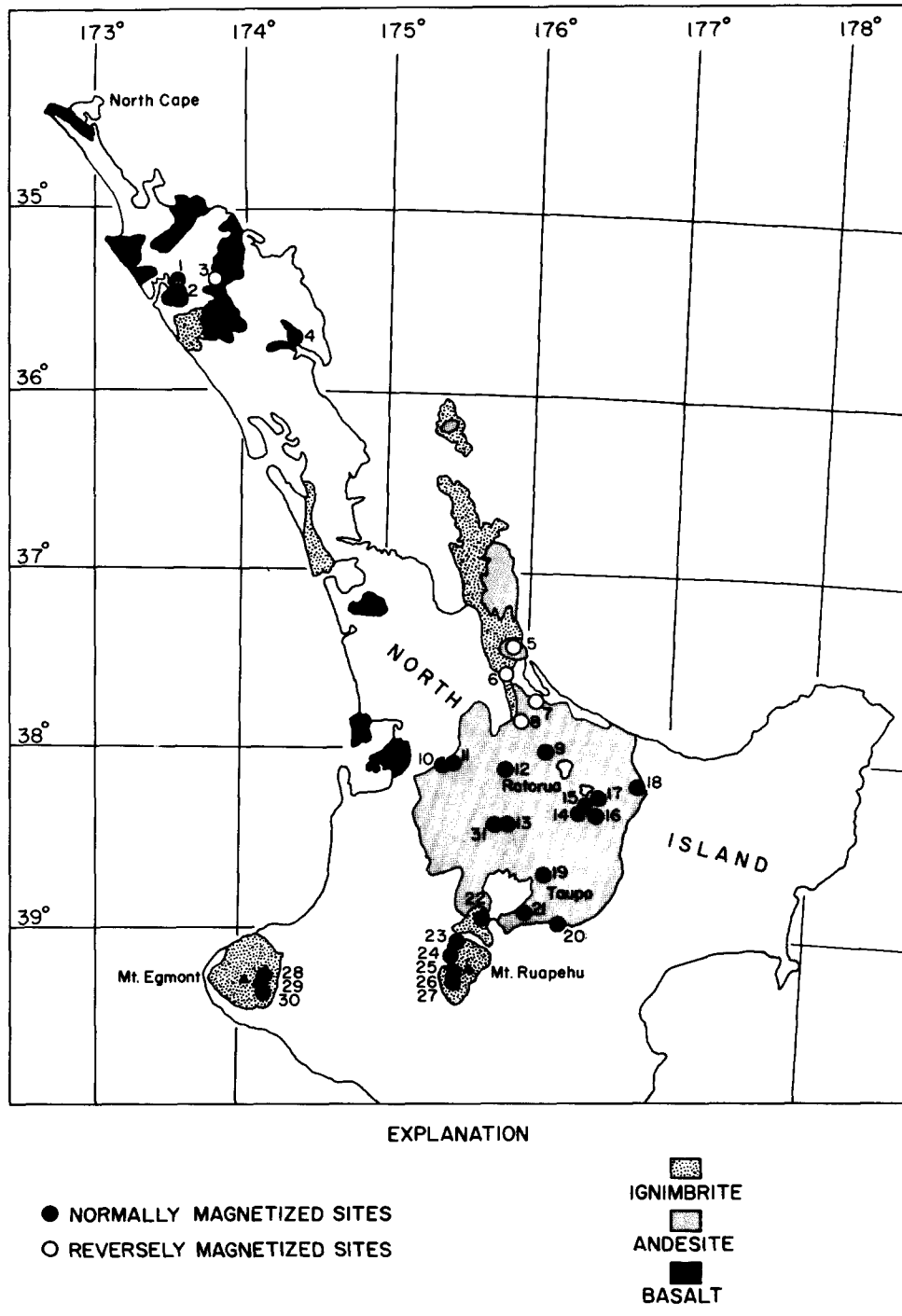



Fig. 1. Sampling sites on North Island, New Zealand.

Table 1
Paleomagnetic stratigraphy.

| | | Andesite flows of Mt. Egmont | Ruapehu and Taupo andesite and dacite | Taupo-Rotorua ignimbrites | Rotorua rhyolites and andesites | North Cape basalts | | |
|---|---|---|---|---|---|---|---|---|
| Holocene | | Normal NZ-28 | | | Normal NZ-17 | | Younger group | Brunhes normal polarity epoch |
| Pleistocene | Hawera Series | Normal NZ-29<br>Normal NZ-30 | Normal NZ-27<br>Normal NZ-26<br>Normal NZ-23<br>Normal NZ-25<br>Normal NZ-22<br>Normal NZ-19<br>Normal NZ-24 | | | Normal NZ-2 | Younger group | Brunhes normal polarity epoch |
| | | | | Normal NZ-16<br>Normal NZ-9<br>Normal NZ-18<br>Normal NZ-13 | | | Older group | Brunhes normal polarity epoch |
| | Wanganui Series | | | Normal NZ-11<br>Normal NZ-21<br>Normal NZ-31<br>Normal NZ-20<br>Normal NZ-15<br>Normal NZ-12<br>Normal NZ-10<br>Normal NZ-14 | | Normal NZ-1<br>Normal NZ-4 | Older group | Brunhes normal polarity epoch |
| | | | | Reversed NZ-8<br>Reversed NZ-5 | Reversed NZ-6<br>Reversed NZ-7 | Reversed NZ-3 | | Matuyama reversed polarity epoch |

## 3. PALEOMAGNETIC RESULTS

Paleomagnetic measurements of two different types were made, depending on the information desired. At eight sites (NZ-1, 3, 5, 6, 7, 8, 10 and 14), rather crude measurements of the magnetic polarity were made for purposes of stratigraphic control. The direction of magnetization was first measured at the outcrop using a portable fluxgate magnetometer [13]. One of the samples was retained and the stability of its magnetization was later checked in the laboratory. In all cases, the laboratory measurements were in accord with the polarity determination made at the outcrop. These results (Appendix), although useful for paleomagnetic stratigraphy, are not precise enough for studies of SV.

For the main analysis of SV, samples were collected from the remaining 22 sites, all in normally magnetized volcanic units in the Brunhes normal epoch (table 2). To achieve a favorable ratio of signal to noise in the measurement of the angular dispersion, eight samples were drilled at each site and oriented

Table 2
Paleomagnetic results from each site.

| Site No. | Site Lat. | Site Long. | $N$ | Incl. | Decl. | J ± st. dev. ($10^{-3}$ gauss) | $R$ | $k$ | $\alpha_{95}$ | VGP Lat. | VGP Long. | δp | δm | Exp. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NZ-2 | −35.46° | 173.65° | 8 | −77.3° | 276.7° | 5.68 ± 2.89 | 7.41354 | 12 | 16.7° | | | | | NRM |
| NZ-2 | −35.46° | 173.65° | 8 | −59.8° | 022.2° | 3.36 ± 1.43 | 7.97705 | 305 | 3.18° | 71.8° | 286.8° | 3.61° | 4.79° | H200 |
| NZ-4 | −35.69° | 174.33° | 8 | −50.2° | 297.8° | 1.72 ± 3.06 | 3.80304 | 1.7 | 65.8° | | | | | NRM |
| NZ-4 | −35.69° | 174.33° | 8 | −81.7° | 349.7° | 0.167 ± 0.143 | 7.93464 | 107 | 5.38° | 51.6° | 358.9° | 10.13° | 10.44° | H200 |
| NZ-9 | −37.97° | 176.16° | 8 | −84.2° | 118.0° | 0.265 ± 0.073 | 7.95071 | 142 | 4.66° | 31.9° | 344.1° | 9.04° | 9.18° | NRM |
| NZ-11 | −38.07° | 175.64° | 8 | −65.2° | 000.6° | 0.161 ± 0.073 | 7.97807 | 319 | 3.10° | 80.8° | 352.9° | 4.06° | 5.01° | NRM |
| NZ-12 | −38.11° | 175.84° | 8 | −63.4° | 343.1° | 1.41 ± 0.25 | 7.99591 | 1713 | 1.34° | 75.7° | 052.0° | 1.67° | 2.12° | NRM |
| NZ-13 | −38.43° | 175.86° | 8 | −61.8° | 009.8° | 0.195 ± 0.035 | 7.97754 | 312 | 3.14° | 81.3° | 300.5° | 3.76° | 4.86° | NRM |
| NZ-15 | −38.38° | 176.33° | 8 | −65.9° | 005.2° | 1.26 ± 0.34 | 7.98537 | 478 | 2.54° | 79.5° | 336.9° | 3.39° | 4.15° | NRM |
| NZ-16 | −39.40° | 176.55° | 8 | −50.6° | 008.2° | 4.17 ± 1.19 | 7.98671 | 527 | 2.42° | 79.5° | 218.6° | 2.19° | 3.26° | NRM |
| NZ-17 | −38.23° | 176.51° | 8 | −62.7° | 011.6° | 0.163 ± 0.31 | 7.94288 | 123 | 5.02° | 79.5° | 304.1° | 6.16° | 7.86° | NRM |
| NZ-18 | −38.30° | 176.78° | 8 | −49.8° | 006.7° | 1.39 ± 0.27 | 7.99310 | 1014 | 1.74° | 80.5° | 214.2° | 1.55° | 2.32° | NRM |
| NZ-19 | −38.68° | 176.17° | 8 | −74.1° | 318.8° | 0.610 ± 0.059 | 7.97908 | 335 | 3.03° | 56.5° | 032.4° | 4.95° | 5.47° | NRM |
| NZ-20 | −38.93° | 176.48° | 8 | −65.9° | 357.4° | 1.30 ± 0.13 | 7.99223 | 901 | 1.85° | 80.6° | 006.9° | 2.47° | 3.02° | NRM |
| NZ-21 | −38.88° | 175.99° | 8 | −64.1° | 357.7° | 0.849 ± 0.121 | 7.98362 | 427 | 2.68° | 82.8° | 009.1° | 3.41° | 4.28° | NRM |
| NZ-22 | −38.99° | 175.77° | 8 | −56.3° | 337.1° | 0.744 ± 0.313 | 7.34652 | 11 | 17.7° | | | | | NRM |
| NZ-22 | −38.99° | 175.77° | 8 | −52.6° | 344.7° | 1.03 ± 0.21 | 7.97804 | 319 | 3.11° | 76.4° | 106.2° | 2.95° | 4.29° | H050 |
| NZ-23 | −39.04° | 175.65° | 8 | −64.0° | 347.8° | 0.703 ± 0.154 | 7.99163 | 837 | 1.92° | 78.8° | 45.0° | 2.44° | 3.06° | NRM |
| NZ-24 | −39.11° | 175.54° | 8 | −53.9° | 012.1° | 4.78 ± 0.79 | 7.99270 | 958 | 1.79° | 79.3° | 243.3° | 1.75° | 2.50° | NRM |
| NZ-25 | −39.23° | 175.55° | 8 | −55.6° | 005.8° | 0.979 ± 0.273 | 7.96160 | 182 | 4.11° | 84.5° | 233.8° | 4.20° | 5.88° | NRM |
| NZ-26 | −39.24° | 175.53° | 8 | −61.9° | 025.1° | 1.16 ± 0.34 | 7.52655 | 15 | 14.9° | | | | | NRM |
| NZ-26 | −39.24° | 175.53° | 8 | −61.5° | 013.8° | 0.687 ± 0.226 | 7.96546 | 203 | 3.90° | 79.0° | 287.9° | 4.63° | 6.01° | H100 |
| NZ-27 | −39.23° | 175.54° | 8 | −56.3° | 006.4° | 4.39 ± 2.83 | 7.99334 | 1051 | 1.71° | 84.4° | 242.1° | 1.78° | 2.46° | NRM |
| NZ-28 | −39.33° | 174.11° | 8 | −65.2° | 347.0° | 11.6 ± 0.6 | 7.99211 | 887 | 1.86° | 77.7° | 040.0° | 2.43° | 3.01° | NRM |
| NZ-29 | −39.29° | 174.09° | 8 | −60.3° | 359.8° | 2.67 ± 1.86 | 7.38644 | 11 | 17.1° | | | | | NRM |
| NZ-29 | −39.29° | 174.09° | 8 | −56.6° | 008.6° | 2.57 ± 1.93 | 7.98737 | 554 | 2.35° | 82.9° | 249.3° | 2.46° | 3.40° | H100 |
| NZ-30 | −39.30° | 174.09° | 8 | −54.7° | 356.0° | 4.59 ± 1.88 | 7.98332 | 420 | 2.71° | 84.8° | 134.6° | 2.71° | 3.83° | NRM |

$N$ : number of samples
$R$ : length of vector sum of $N$ unit vectors
$k$ : precision parameter
$\alpha_{95}$ : 95% confidence circle
δp, δm: confidence limits on VGP (virtual pole)
Exp. : NRM, original magnetization; H100, magnetization after cleaning in field of 100 Oe.

using geographic coordinates. The remanent magnetism was measured using a spinner magnetometer [19]. At least one specimen from each site was partially demagnetized in alternating magnetic fields of 12.5, 25, 50, 100, 200, 400, and 800 oersteds. The direction of magnetization generally did not change if the original directions of magnetization of all samples from a site were well-grouped. For sites where this was true, the original directions of magnetization were used to find the mean direction because magnetic cleaning neither changed the direction of the mean nor improved its accuracy. Magnetic cleaning was used for sites with scattered directions of magnetization and for sites for which the test specimen underwent a change in direction of magnetization on partial demagnetization. The decay of the natural remanence in small alternating fields varied considerably from specimen to specimen, depending on the amount of unstable magnetization present. The decay in large alternating fields was typical of that of volcanic rocks known to have cooled in the earth's field [20]. The mean direction at each site (table 2) may therefore reasonably be regarded as the direction that existed at the site when the flow originally cooled, subject to experimental errors as discussed below.

The volcanic units were divided into two groups of nearly equal size, using the top of the ignimbrite sequence as an easily traceable boundary. The angular dispersion appears to be slightly greater among the older sites (fig. 2). To test whether the difference between older and younger sites is significant, the ratio of the precision parameters of the two groups was calculated (table 3) and found to be 1.84, whereas the critical $F^{=}$ ratio at 95% confidence for 18° and 22° of freedom is 2.43. There is thus no basis for rejecting the null hypothesis that the difference is due to random sampling effects. The present data do not indicate that SV decreased during the Brunhes polarity epoch.

The quantity of principal interest in the present study is angular dispersion of site mean directions of magnetization, as described by the precision parameter $k_T = 49$. The total dispersion consists of two components, between-site dispersion ($k_B$) due mainly to ancient geomagnetic secular variation, and within-site dispersion ($k_W$), which for present purposes is noise. The value of $k_B$ may be found in either of two ways. Standard analysis of variance yields the rela-

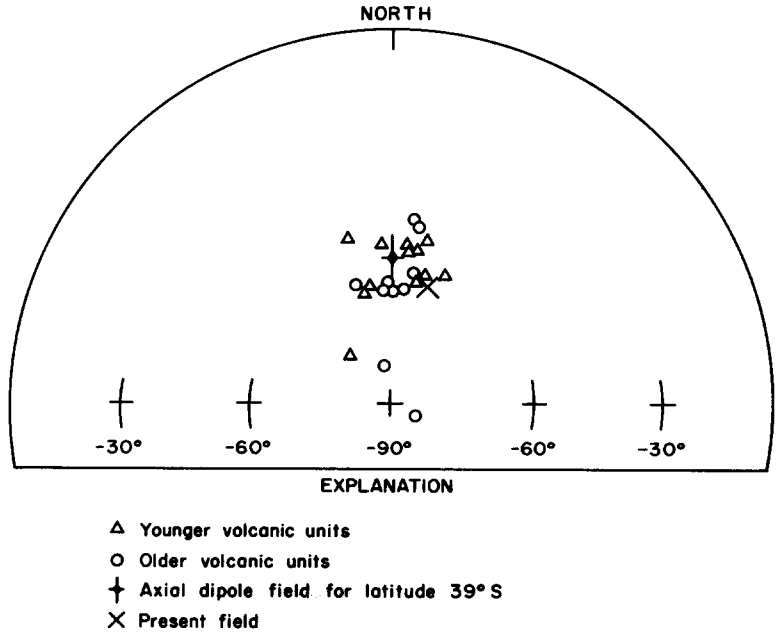



Fig. 2. Average direction of magnetization for each site. Older volcanic units include all of the ignimbrites and two basalts from North Cape.

tionship

$$\frac{1}{k_T} = \frac{1}{k_B} + \frac{1}{Nk_W}, \quad (1)$$

where $N = 8$ is the number of samples at each site. The average value of the within-flow dispersion expressed as angular standard deviation is 4.22°, corresponding to $k_W = (81°/4.22°)^2 = 370$. From eq. (1), $k_B = 49.6$. Alternatively, if the two-level analysis of dispersion of Watson and Irving [21] is used, values are found of $k_W = 322$ and 49.9, in good accord with the other analysis.

An alternative measure of dispersion is the angular standard deviation from the mean calculated directly using angles (table 3). Again the total dispersion $s_T$ may be separated into parts, one due to within-site variation and one to between-site variation:

$$s_T^2 = s_B^2 + s_W^2/N. \quad (2)$$

If a value of $s_W = 4.22$ is used corresponding to $k_W = 370$, the value of $s_B = 11.56°$. If the alternative value of $s_W = 4.52°$ is used corresponding to $k_W = 322$, the value of $s_B = 11.55°$. The small difference between the two values of $s_B$ indicates that the value of the between-site dispersion is much greater than the noise due to within-site dispersion. Therefore it is insensitive to small uncertainties in the magnitude of the within-site dispersion.

Although the between-site dispersion is mainly the result of ancient secular variation of the main field, it also includes a small component due to local magnetic anomalies that existed at the sampling site when the flows originally cooled. Unfortunately, the local anomalies at the sampling sites today cannot be used to correct the ancient field directions because the magnetic topography responsible for the anomalies has changed since the flows formed. However, an estimate can be made of the average angular standard deviation $s_A$ that existed at the sampling sites on the basis of the local angular variations that exist over similar magnetic topography today. From the results of an earlier study of this type [20], an estimate of $s_A = 1.25$ is reasonable for the angular standard deviation on New Zealand. If $s_B$, $s_F$, and $s_A$ are the angular standard deviations of between-site dispersion, ancient main field secular variation, and dispersion due to local anomalies, respectively,

$$s_F^2 = s_B^2 - s_A^2. \quad (3)$$

Using the value of the between-site standard deviation from the mean $s_B = 11.56°$, the best estimate of the ancient angular SV is $s_F = 11.49°$ with 95% confidence limits (42° of freedom) between 9.5° and 14.4°.

Implicit in these calculations was the assumption that the average direction of the field during the Brunhes epoch is unknown and is best estimated by the vector sum of the 22 New Zealand site mean directions. Slightly larger values ($s'_T$ in table 3) are obtained if the alternative assumption is made that the average field direction was that of an axial dipole; considerable (but not unanimous) support for this may be

Table 3
Statistics for the average directions of magnetization at each site.

| | Younger group | Older group | All |
|---|---|---|---|
| Mean inclination | −60.6° | −66.2° | −63.1° |
| Mean declination | 1.5° | 2.7° | 2.0° |
| $N$ | 12 | 10 | 22 |
| $R$ | 11.8395 | 9.7566 | 21.5696 |
| $k_T$ | 69 | 37 | 49 |
| $\alpha_{95}$ | 5.28° | 8.05° | 4.48° |
| $s_T$ | 9.80° | 13.40° | 11.66° |
| $s'_T$ | 10.34° | 16.78° | 13.30° |

$N$ : number of sites,

$k_T$ : precision paramater, $\frac{N-1}{N-R}$,

$\alpha_{95}$: circle of confidence,

$s_T$ : $\left(\frac{1}{N-1}\sum_{i=1}^{N}\delta_i^2\right)^{\frac{1}{2}}$,

$\delta_i$ : angle between direction at $i$th site and overall mean,

$s'_T$ : $\left(\frac{1}{N-1}\sum_{i=1}^{N}(\delta'_i)^2\right)^{\frac{1}{2}}$,

$\delta'_i$ : angle between direction at $i$th site and axial dipole field at $i$th site.

Mean virtual geomagnetic pole for all sites: latitude, 82.5°N, longitude, 346.2°E, $\alpha_{95}$, 6.9°.

found in earlier paleomagnetic studies [9, 23, 24]. On correcting the value for $s'_T$ for small contributions due to within-flow variation and ancient anomalies, the best estimate of the ancient angular deviation from the field of an axial dipole is $s'_F = 13.23°$ with 95% confidence limits (44 degrees of freedom) between 10.9° and 16.7°. The range of this confidence limit is much larger than the difference between the values of angular deviation from the mean direction (11.49°) and from the direction of the axial dipole field (13.23°) and is two orders of magnitude larger than corrections for within-site scatter and ancient magnetic anomalies. The overall accuracy of the present measurement of angular deviation is limited mainly by the intrinsic statistical uncertainty of the experiment, which depends on the number of sample sites.

Table 4
Statistical analysis of virtual geomagnetic poles.

| | A | B | C |
|---|---|---|---|
| $N$ | 22 | | |
| $R$ | 21.0071 | | |
| $\alpha_{95}$ | 6.9° | | |
| $K$ | 21.2 | 21.4 | 21.5 |
| $K_0$ | 19.5 | 19.6 | 19.7 |
| $S$ | 17.90° | 17.83° | 17.79° |
| $S_0$ | 19.65° | 19.59° | 19.55° |

Column A: Uncorrected values.
Column B: Corrected for within-site dispersion.
Column C: Corrected for within-site dispersion and for local anomalies.
$K$, $S$ : Estimates of the precision parameter and angular standard deviation about the average pole.
$K_0$, $S_0$ : Estimates of the precision parameter and angular standard deviation about the present geographic pole.

The 95% confidence limit about the mean virtual geomagnetic pole for these results (column A, table 4) excludes the present geographic pole. In a similar study of lava flows from the Galapagos Islands and the western United States including Hawaii and Alaska, Cox and Doell [24] found that the mean pole for 211 sites was significantly different from the present geographic pole, suggesting that the earth's rotation axis may have been displaced slightly during the Brunhes epoch. The present New Zealand mean pole and the earlier one are mutually consistent, but the support supplied by the present study is weak because the number of sample sites is small.

The angular dispersion of the 22 virtual geomagnetic poles corresponding to the 22 site mean directions (column A, table 4) was corrected for between-site dispersion (column B) and local magnetic anomalies (column C). The latter provides a measure of ancient SV expressed in terms of poles. The small magnitude of the corrections indicates that the signal due to ancient SV is well above the level of experimental noise.

## 4. DISCUSSION

The angular dispersion of virtual poles in New Zealand is greater than that at nearly the same latitude in North America (fig. 3). However, the rather broad confidence limits for the New Zealand results (16.2° to 24.7°) suggests the difference is not significant.

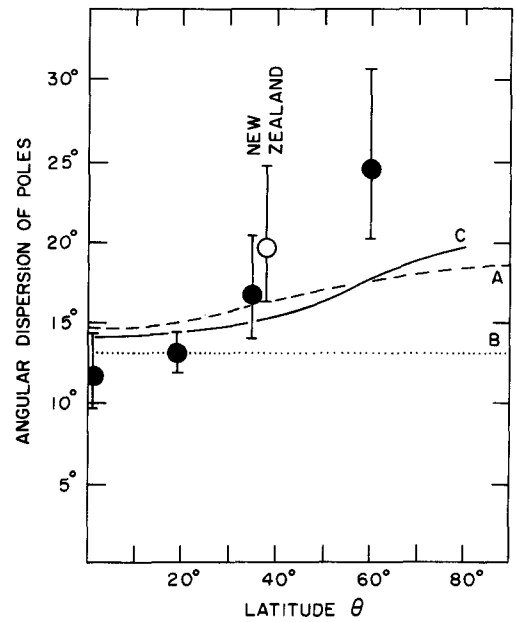



Fig. 3. Angular dispersion of virtual poles from Galapagos Islands ($\theta$ = 0°), Hawaii ($\theta$ = 19°), North America ($\theta$ = 37°), New Zealand ($\theta$ = 38°), and Alaska ($\theta$ = 60°). The models for curves A[8], B [7], and C [2] are described in the text.

This is confirmed by an F-test: the ratio of values of $K$ for the two hemispheres is 1.17, whereas at 95% confidence $F_{44,54} = 1.76$. Thus the present results do not prove that the present asymmetry between the hemispheres has also existed in the past.

Three models have been proposed to describe how SV may have varied with latitude in the past. *Model A* [8]: SV is produced by adding vectors of constant amplitude and varying orientation to the field of a constant axial dipole. The latitude dependence is the result of the increase of the dipole field with latitude. *Model B* [7]: SV is produced entirely by the wobble of a geocentric dipole, the nondipole field being neglected. The angular dispersion of virtual poles does not vary with latitude (curve B of fig. 3). *Model C* [2]: SV is due in part to a wobble of $11\frac{1}{2}°$ of a geocentric dipole and in part to variations of the nondipole field. The latitude dependence of the latter is estimated from the present nondipole field. For curve A in fig. 3, results obtained for the present field along circles of latitude in the northern and southern hemispheres were averaged.

The angular dispersion of virtual poles in New Zealand is greater than that calculated by any of the models for a latitude of 38°. The chi-square statistic $2NS_0^2/S_P^2$ may be used to test the significance of the difference, $N$ being the number of sample sites, $S_0$ the observed angular standard deviation of virtual poles, and $S_P$ the predicted value. Using the value $S_P = 16.2$ for model A, which has the largest angular dispersion at this latitude of any of the models, a chi-square value of 64.4 is found, whereas at 95% confidence $\chi^2_{44} = 64.2$. The observed angular dispersion is thus significantly different from that predicted by model A, but just barely so. The departure of the New Zealand results from the values predicted by models B and C is even greater. The present study, together with the earlier results from Alaska [24], indicates that in the past, the amplitude of secular variation at high latitudes was greater than predicted by present models.

## APPENDIX

(Lower case letters such as om and mr refer to geologic map units in references [10] and [11]. Numbers such as N14-107367 are New Zealand national reference grid coordinates.)

| *Site* | |
|---|---|
| NZ-1 | Horeke Basalt (?) (N14-107367). Sampled along Taheke-Horeke Road 0.7 miles southeast of McDonnel Road. Normally magnetized. |
| NZ-2 | Taheke Basalt (N14-163280). Cut on north side of main highway. |
| NZ-3 | Horeke Basalt (N15-337305). Sampled in quarry 2 miles south of Kaikohe on the east side of the airport road. Reversely magnetized. |
| NZ-4 | Horeke Basalt (N20-847016). Streambed at Whanarei Falls. |
| NZ-5 | Unmapped ignimbrite (N53-268927). Flow at least 15 meters thick with massive columnar joints sampled in outcrop on west side of Waitekauri Road near railroad crossing 3.9 miles south of Waihi on Paeroa-Waihi Highway. Reversely magnetized. |
| NZ-6 | Omahia Andesite (om [11]) (N53-414861). Quarry at junction of Waihu-Tau range road with Athenree Road. Reversely magnetized. |
| NZ-7 | Minden Rhyolite (mr [11]) (N58-526568). East side of the Minden Quarry. Reversely magnetized. |
| NZ-8 | Waiteariki Ignimbrite (N67-526448). Sampled in bed of river at McLarens Falls. Mapped as Mamaku ignimbrite [11] but J.Healy (personal communication) states that it is probably a much older unit, Waiteariki Ignimbrite. Reversely magnetized. |
| NZ-9 | Mamaku Ignimbrite (mk [11]) (37.97°S, 176.16°E). Quarry facing road between Chaupara and Mangarewa Streams 300 meters above their confluence. |
| NZ-10 | Ahuroa Ignimbrite member of Pakaumanu Group (paa [11]) (N75-021119). Sampled in roadcut 2.3 miles west of Pukeatua. Normally magnetized but rather unstable. |
| NZ-11 | Manunui Ignimbrite (mn [11]) (N75-138130). Cut in road south of Arapuni Power Station. |
| NZ-12 | Marshall Ignimbrite (ms [11]) (N75-323081). Small quarry on south side of road about 1 mile east of Lichfield. |
| NZ-13 | Haparangi Rhyolite ($ha_1$ [11]) (N84-342694). Cliff outcrop along road on north side of Lake Whakamaru. |
| NZ-14 | Te Kopia Ignimbrite (te [11]) (N85-733766). Sampled at base of precipitous rocky face forming a prominent fault scarp. Normal polarity. |
| NZ-15 | Waiotapu Ignimbrite (wp [11]) (N85-799761). Cliff outcrop sampled near base. Trend of columnar joints suggests dip of 10° in azimuth 075°, but correction not made because of possibility that ignimbrite cooled against a sloping surface. |
| NZ-16 | Kaingaroa Ignimbrite ($kg_2$ [11]) (86-998720). Forrestry Quarry on Webb Road. Samples collected near the middle of exposure which is 30 meters thick. |
| NZ-17 | Tarawera volcanic complex [25] (N77-962928). Extreme western part of Ruawahia Dome as ex- |

posed on the north wall of the rift immediately east of the divide southeast of the Tarawera Trig Point.

NZ-18 Matahina Ignimbrite (ma [11]) (N86-222861). Shallow cut on Te Teko Murupara Road.

NZ-19 Tauhara Dacite (ta [10]) (N94-638384). Mt. Tauhara, quarry exposing dike leading to the west rim of a small cone on the last flank of the main volcano.

NZ-20 The Whaiti Ignimbrite (tw [10]) (38.93°S, 176.48°E). Cut in Napier-Taupo Road 3.0 miles west of Waipunga Stream.

NZ-21 Rangitaiki Ignimbrite (ra [10]) (N103-464144). Small cliff 100 meters east of road adjacent to Hallets Bay, Lake Taupo.

NZ-22 Pihanga Andesite (pi [10]) (N102-260008). Cut in road of Ministry of Works west of Papanetu Stream.

NZ-23 North Crater Andesite (nc [10]) (N12-140956). Large quarry northwest of Otukou Village.

NZ-24 Tongariro Andesite (to [10]) (39°06.41′S, 175°32.4′E). Exposures in cut on Taupa State Highway 3 miles south of Taurewa.

NZ-25 Ruapehu Andesite (ru [10]) (39°41.1′S, 175°33.1′E). Near base of series of thick flows which make up Te Herenga Ridge.

NZ-26 Ruapehu Andesite (ru [10]) (39°14.5′S, 175°31.9′E). 1200 yards south 45 west of Salt Hut.

NZ-27 Ruapehu Andesite (ru 1 [10]) (39°14.0′S, 175°32.4′E). Along Waikare Streambed.

NZ-28 Andesite flow of Mt. Egmont (N119-688592). Streambed at Dawson's Falls.

NZ-29 Andesite flow of Mt. Egmont (N119-667628). Second highest flow of series forming ridge below Warwick Castle.

NZ-30 Andesite flow of Mt. Egmont (N119-669619). East slope of Mt. Egmont along trail south of ski area on north wall of Monganui Gorge.

NZ-31 Whakamaru Ignimbrite. Normal magnetization reported [26] at three sites along the Waikato River: Waipapa, Maraetai, and the Whakamaru dam site. Not resampled for the present study.

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